THE UNIVERSITY OF TORONTO
DALLA LANA SCHOOL OF PUBLIC HEALTH

Practicum Presentation

From Accelerometer Time Series to Functional Activity Features

Occupation Time Curves, AUC Construction, and Correlation Diagnostics

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Instructor: Osvaldo Espin Garcia

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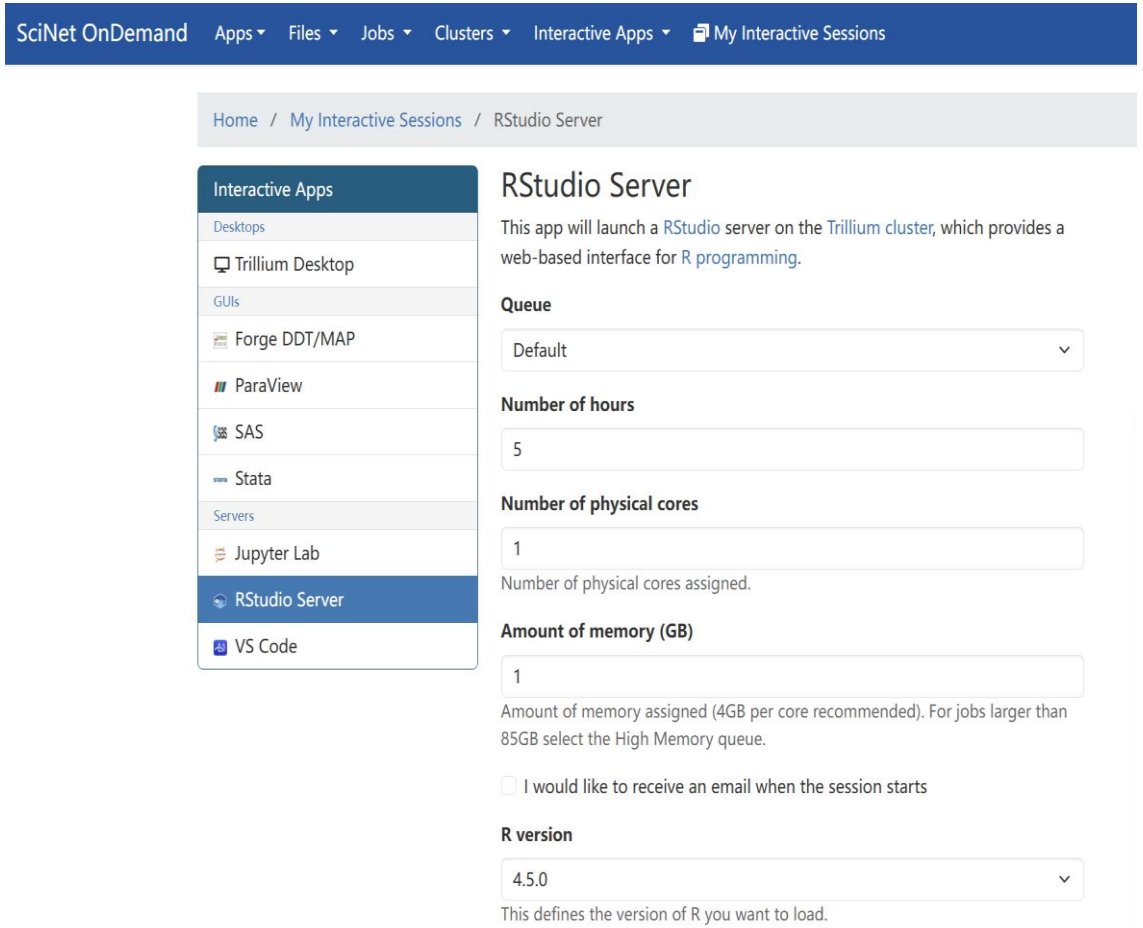
Computing Environment and Remote Access



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Dataset Overview

01 Data Source

- Osteoarthritis Initiative (OAI)
- Large, longitudinal public health study
- Data collected in the United States

02 Accelerometer Data

- Wearable accelerometer measurements
- Minute-level physical activity counts
- Multiple days per participant

Variable Table

Variable	Type	**Description**
ID	Char	**Same ID: same individual**
V08PAStudayDay	Num	**Study day (range 781–1427, with Day 1 = August 01, 2008)**
V08MinSequence	Num	**Time index (t): (range 1–1440, representing minutes from 00:00 to 23:59)**
V08SupectMInute	Num	**Indicator of suspicious minute suggesting monitor wear during sleep**
V08MINCnt	Num	**The core variable, which is the activity count**

The primary variables used in subsequent analyses are **V08MinSequence** (time index) and **V08MINCnt** (activity count), which together form the minute-level activity time series for each individual.

Overall Process

Current Stage

Raw Data → VM(t) → OTC → AUC → Fused Lasso

AUC → L0 + MIO

Raw Data

Minute-level accelerometry data collected for each individual.

Raw activity counts recorded every minute across multiple study days.

VM(t)

Activity time series for a given individual-day.

For each individual and study day, activity counts are represented as a time series $VM(t),\ t = 1, \ldots, 1440$.

OTC

Occupation Time Curve summarizing activity intensity distribution.

OTC captures the proportion of time the activity level exceeds a given threshold, discarding temporal ordering

AUC

Scalar summary of the OTC.

The area under the OTC provides a low-dimensional summary of overall activity intensity.

L0 + MIO

Sparse variable selection via L0-regularized optimization.

Mixed-integer optimization is used to identify important activity features.

Fused Lasso

Structure-aware regression on functional summaries.

Fused lasso encourages sparsity and smoothness across adjacent features.

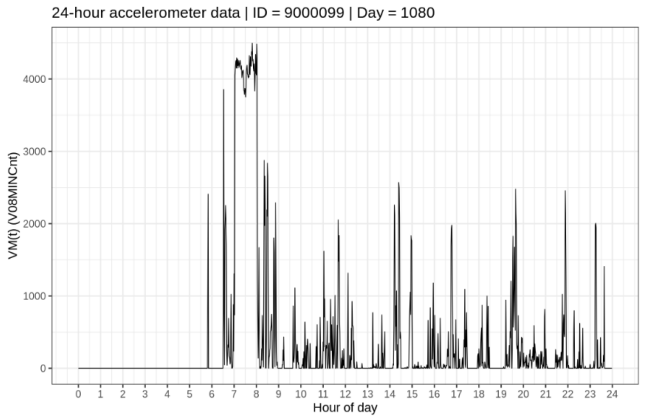
Part 3： Current Step

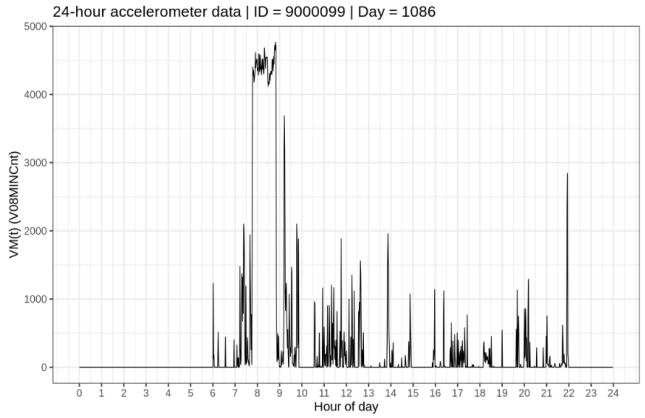
Build VM(t)

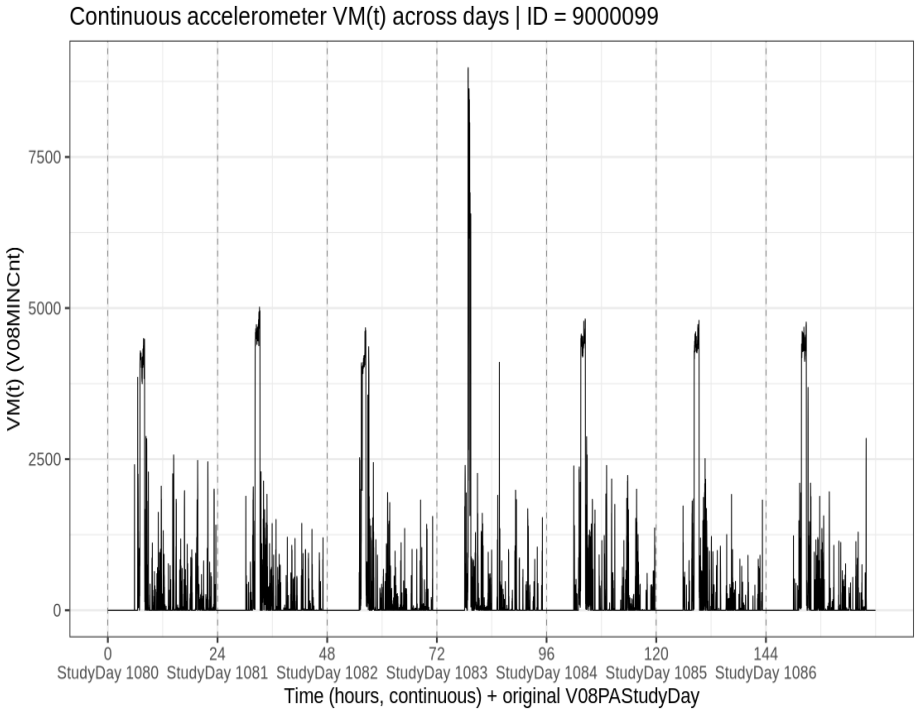
Vector Magnitude time series,

VM(t): minute-level activity count at time t.

The plots on the left show activity within individual days, while the plot on the right combines multiple days into a single continuous timeline.







Why did this?

Raw data: single continuous time axis.

This step: time re-indexing.

Part 3： Current Step

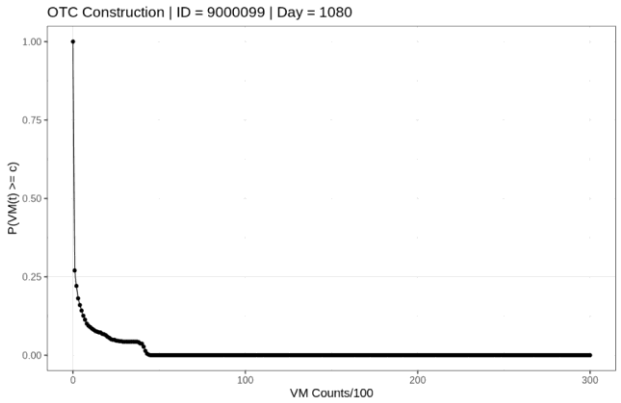
Transform to OTC

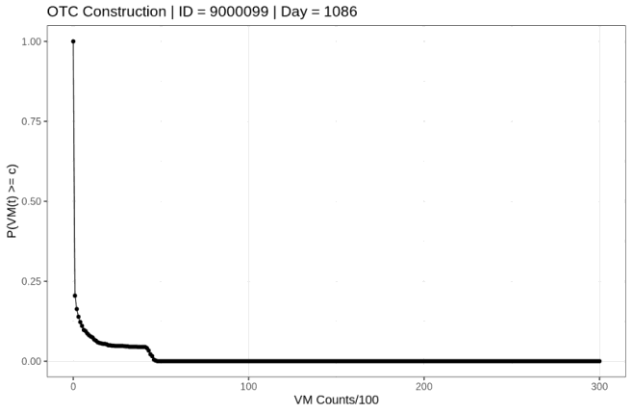
From minute-level activity time series:

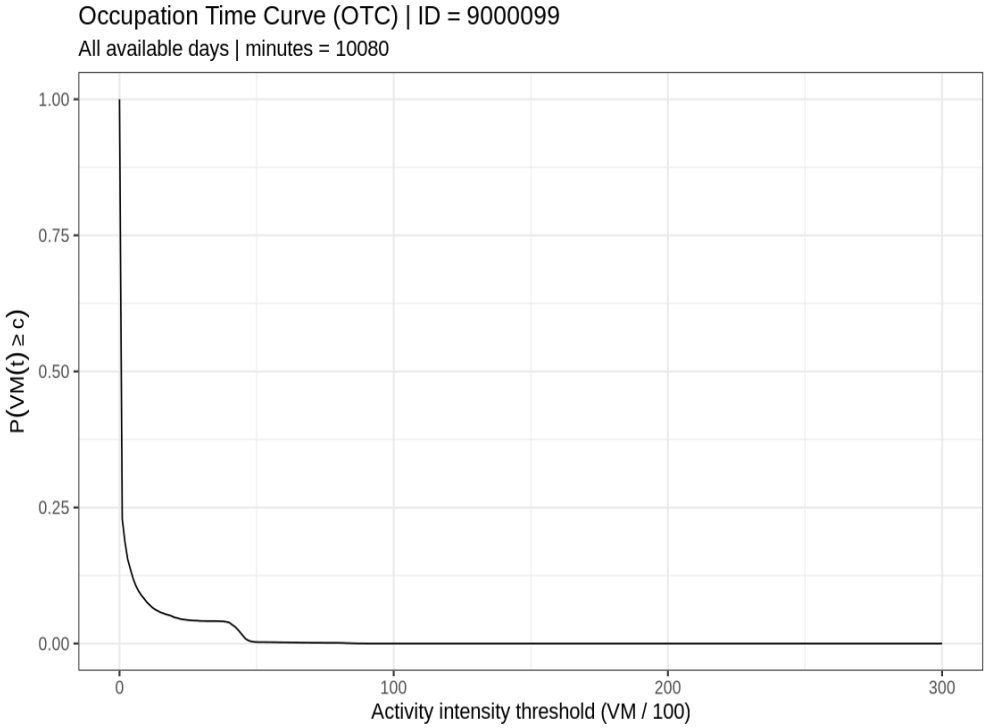
$$VM(t), t = 1, \dots, T$$

to a distributional summary: Occupation Time Curve (OTC):

$$OTC(c) = P\{VM(t) \geq c\}$$





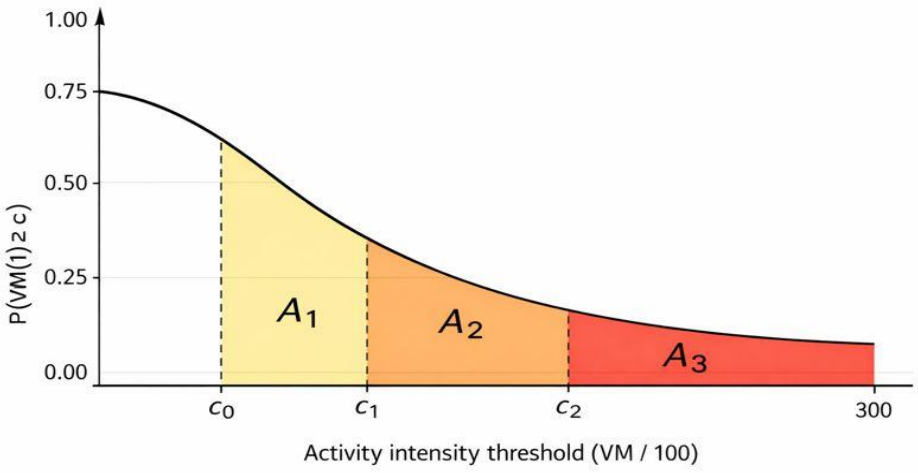


Why did this?

The (OTC) intentionally ignores the temporal ordering of VM(t).

Focuses on a single question: Over the entire observation period**, what proportion of time does an individual spend at or above a given activity intensity level?**

Establish AUC



We partition the activity intensity axis into intervals: $0 = c_0 < c_1 < \cdots < c_J$, and define, for each interval, $A_j = \int_{c_{j-1}}^{c_j} OTC(c)\, dc, j = 1, \ldots, J$.

Example to use AUC:
Outcome Y_i: BMI of subject i
Covariates (from activity data): $(A_{i1}, A_{i2}, \ldots, A_{iJ})$, where: A_{ij}= AUC for subject i in intensity interval j, and fit the model: $Y_i = \beta_0 + \sum_{j=1}^{J} \beta_j A_{ij} + \varepsilon_i$.

If $\beta_j > 0$: Spending more time in intensity range j is associated with higher BMI.
If $\beta_j < 0$: Spending more time in intensity range j is associated with lower BMI

Why did this?

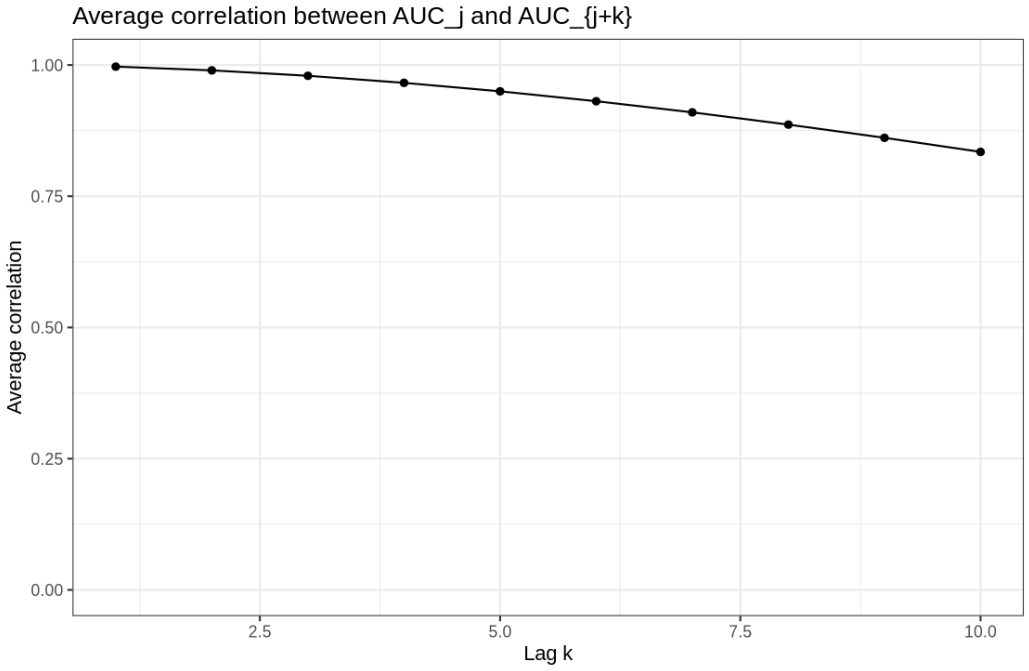
By splitting the OTC into shaded areas, we convert a curve into a set of numerical features.

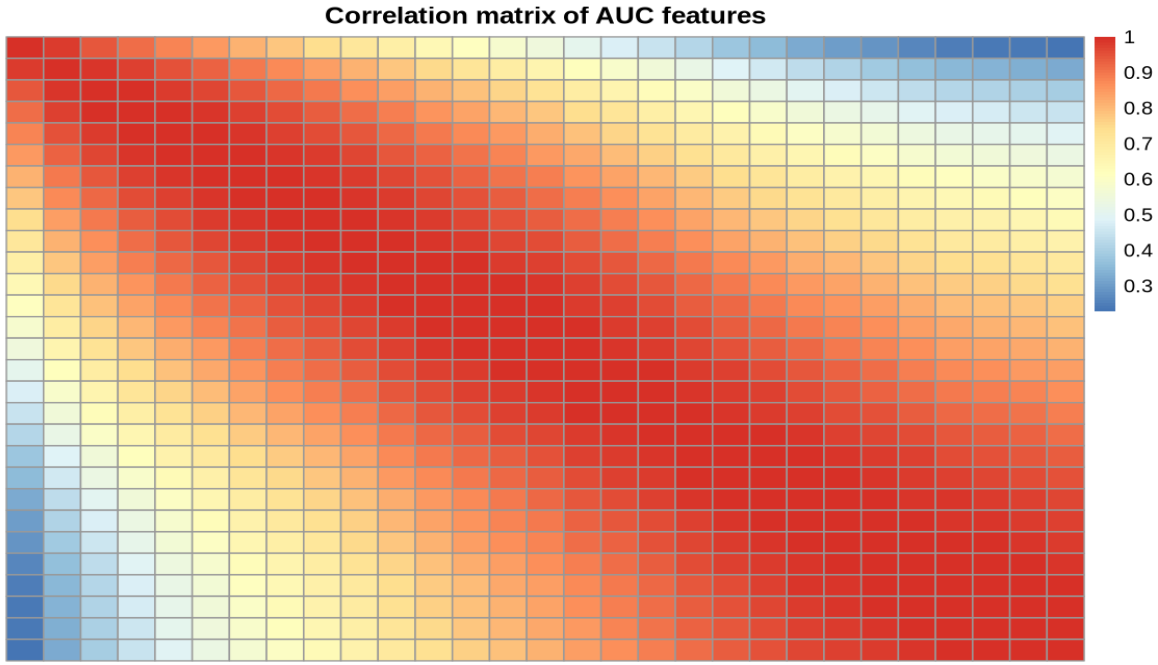
OTC provides a stable functional summary.

AUC converts it into a model-ready representation.

Correlation Diagnostics

Here we check how similar the AUC features are to each other.





Strong correlation alone is not sufficient justification to apply fused lasso or standard regression at this stage.

L0 regularization directly controls how many features are selected, and **mixed-integer optimization** allows us to solve this selection problem exactly.

Plans for next steps

1. Identify an appropriate outcome variable (Y)
2. Apply L0 + MIO for feature selection
3. Examine and interpret selected features

Thanks For Listening

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